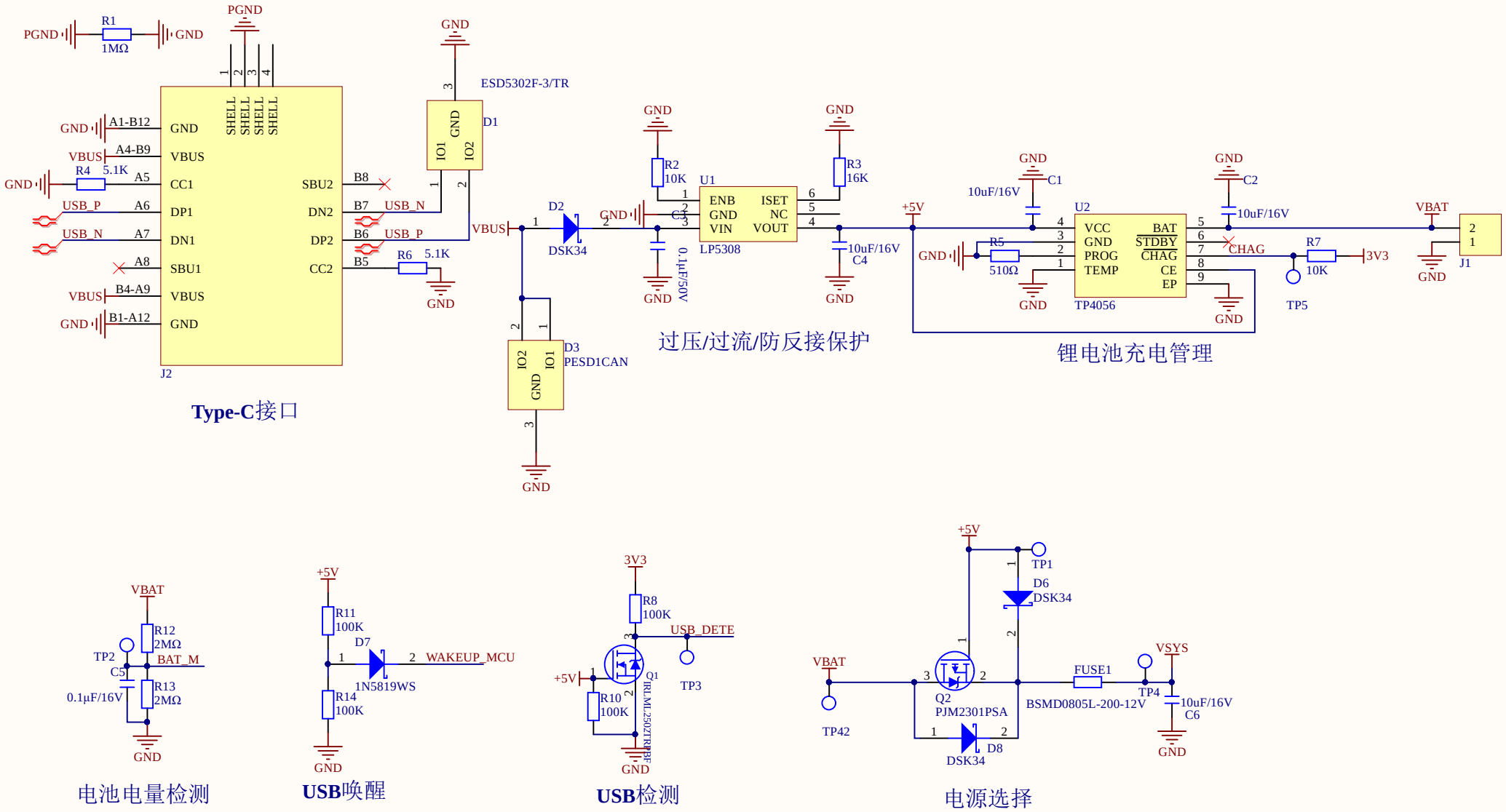
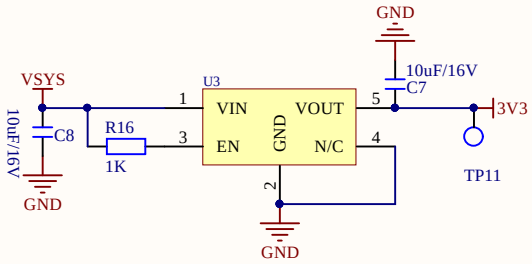
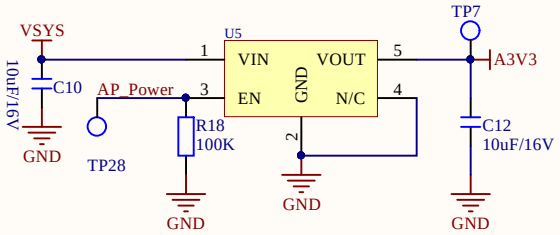

J3
1
3V3
2
ON/OFF
3
Vol-
4
Stop
5
Vol+
6
WAKEUP_MCU
7
GND
8
TP12
TP27
TP33
TP34
TP35
8PIN

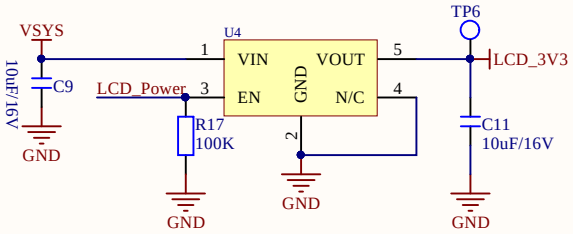

PGND
R1
1MΩ
GND
SHELL
SHELL
SHELL
SHELL
GND
A1-B12
VBUS
A4-B9
R4 5.1K
A5
CC1
USB_P
A6
DP1
USB_N
A7
DN1
A8
SBU1
VBUS
B4-A9
VBUS
GND
B1-A12
GND
SBU2
B8
DN2
B7
USB_N
DP2
B6
USB_P
CC2
B5
R6 5.1K
J2
Type-C接口
ESD5302F-3/TR
D1
IO1
GND
IO2
VBUS
D2
DSK34
D3
PESD1CAN
R2
10K
U1
ENB
GND
VIN
ISET
NC
VOUT
LP5308
R3
16K
C3
0.1uF/50V
C4
10uF/16V
过压/过流/防反接保护
+5V
C1
10uF/16V
U2
VCC
GND
PROG
TEMP
BAT
STDBY
CHAG
CE
EP
TP4056
R5
510Ω
C2
10uF/16V
CHAG
R7
10K
3V3
TP5
VBAT
J1
锂电池充电管理
TP2
C5
0.1μF/16V
R12
2MΩ
BAT_M
R13
2MΩ
电池电量检测
R11
100K
D7
1N5819WS
WAKEUP_MCU
R14
100K
USB唤醒
3V3
R8
100K
USB_DETE
Q1
IRLML2502TRPBF
R10
100K
TP3
USB检测
TP1
D6
DSK34
TP42
Q2
PJM2301PSA
FUSE1
BSMD0805L-200-12V
TP4
VSYS
C6
10uF/16V
D8
DSK34
电源选择



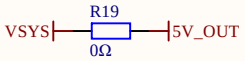
MCU供电



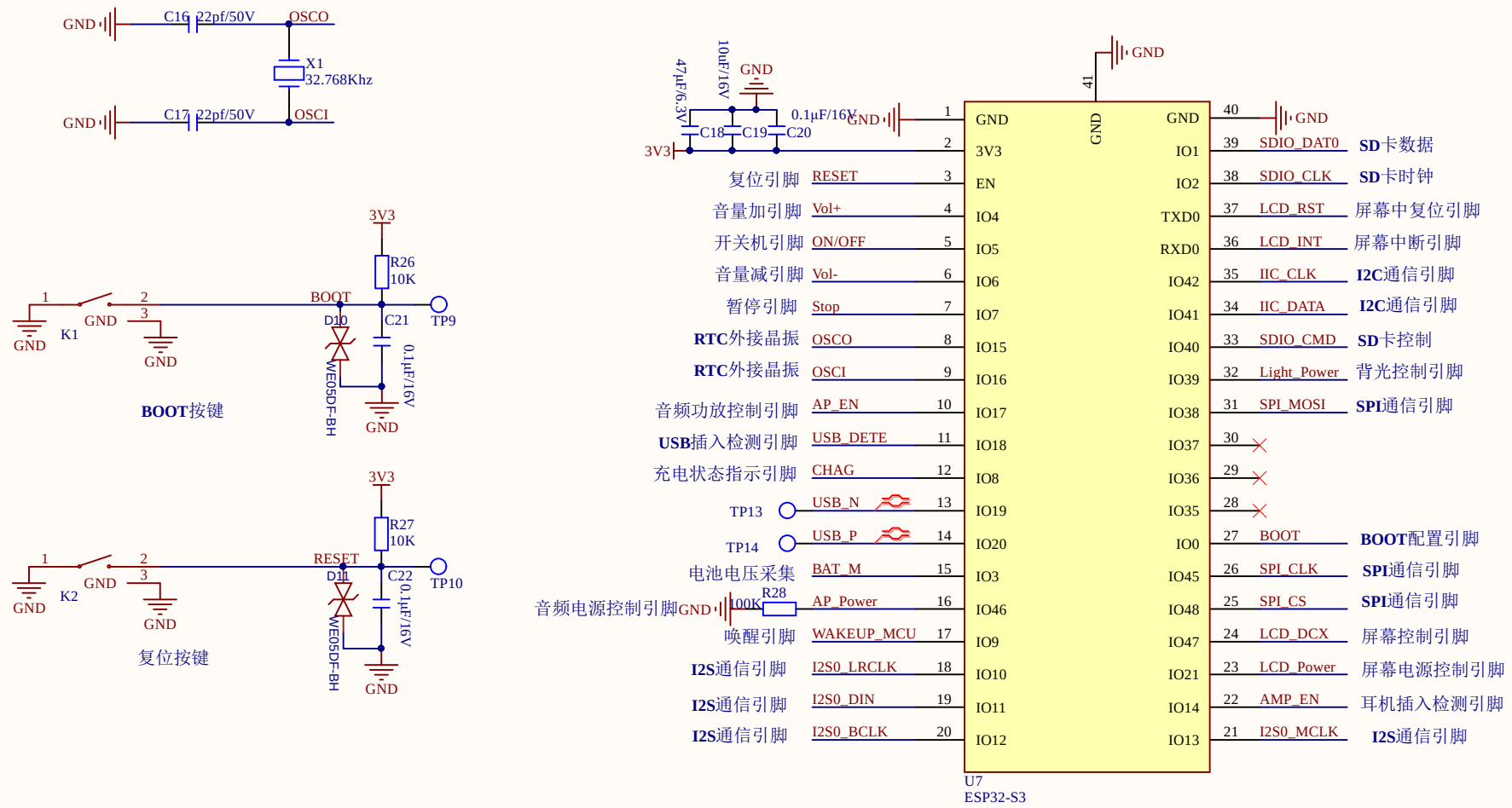
音频芯片供电

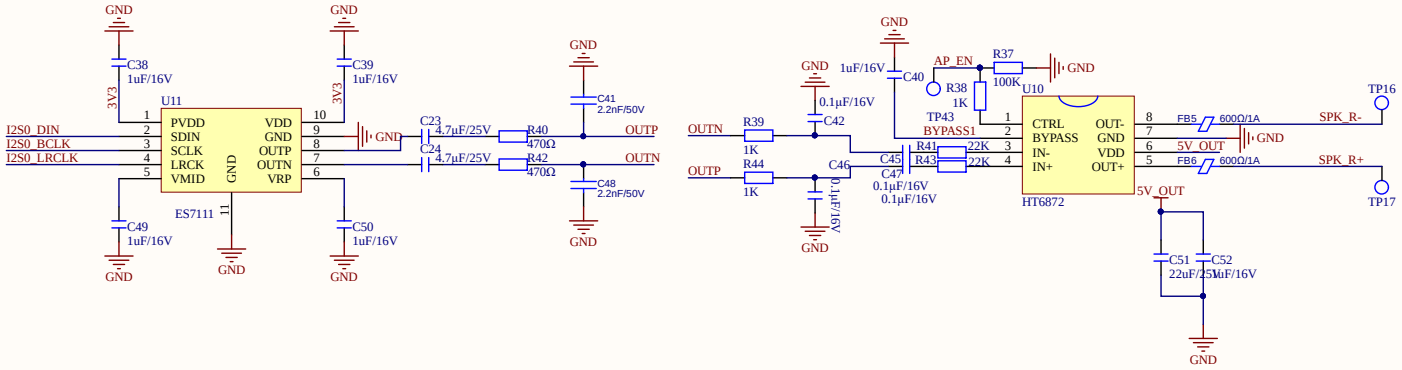


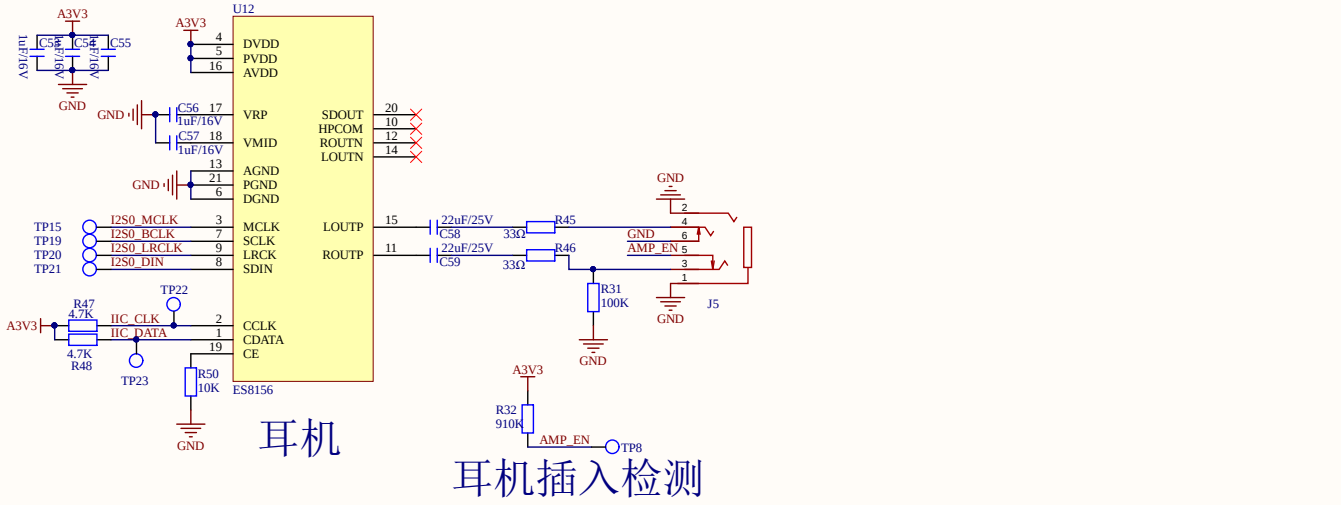
LCD屏幕供电

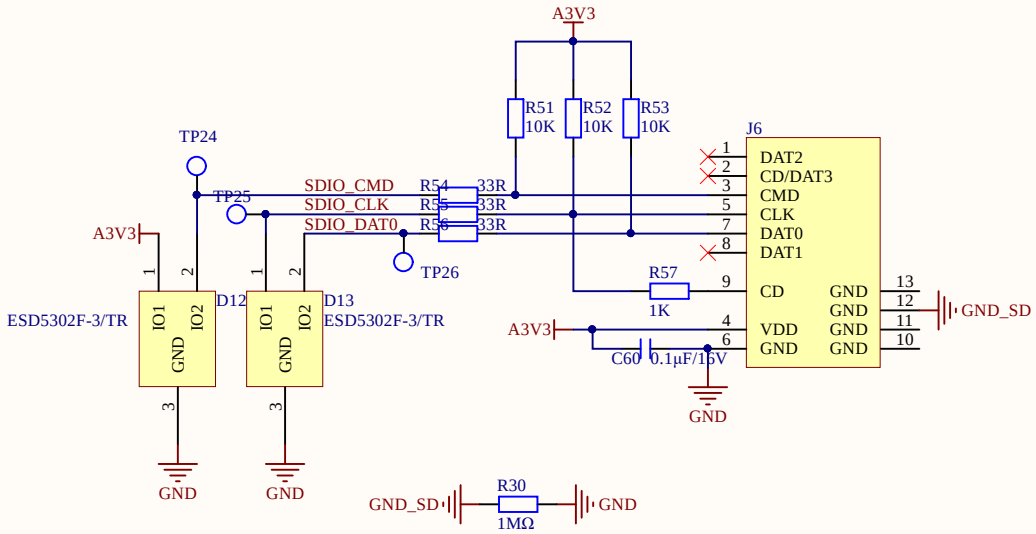


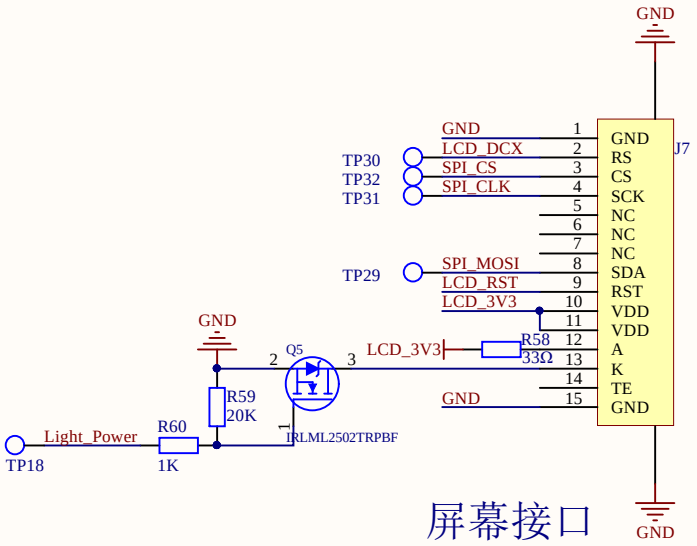
音频功放供电

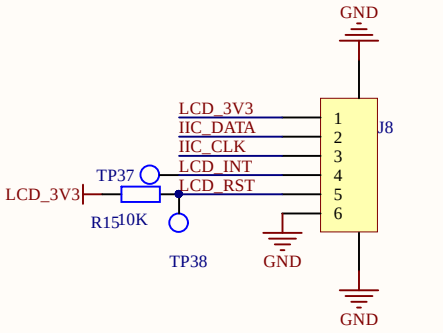

GND
C16 22pf/50V
OSCO
X1
32.768Khz
GND
C17 22pf/50V
OSCI
3V3
R26
10K
1
2
3
BOOT
D10
C21
TP9
K1
GND
WE05DF-BH
0.1μF/16V
BOOT按键
3V3
R27
10K
RESET
D11
C22
TP10
K2
复位按键
47μF/6.3V
10μF/16V
C18
C19
C20
0.1μF/16V
41
U7
ESP32-S3
1 GND
2 3V3
3 EN 复位引脚 RESET
4 IO4 音量加引脚 Vol+
5 IO5 开关机引脚 ON/OFF
6 IO6 音量减引脚 Vol-
7 IO7 暂停引脚 Stop
8 IO15 RTC外接晶振 OSCO
9 IO16 RTC外接晶振 OSCI
10 IO17 音频功放控制引脚 AP_EN
11 IO18 USB插入检测引脚 USB_DETE
12 IO8 充电状态指示引脚 CHAG
13 IO19 TP13 USB_N
14 IO20 TP14 USB_P
15 IO3 电池电压采集 BAT_M
16 IO46 音频电源控制引脚 R28 100K AP_Power
17 IO9 唤醒引脚 WAKEUP_MCU
18 IO10 I2S通信引脚 I2S0_LRCLK
19 IO11 I2S通信引脚 I2S0_DIN
20 IO12 I2S通信引脚 I2S0_BCLK
40 GND
39 IO1 SDIO_DAT0 SD卡数据
38 IO2 SDIO_CLK SD卡时钟
37 TXD0 LCD_RST 屏幕中复位引脚
36 RXD0 LCD_INT 屏幕中断引脚
35 IO42 IIC_CLK I2C通信引脚
34 IO41 IIC_DATA I2C通信引脚
33 IO40 SDIO_CMD SD卡控制
32 IO39 Light_Power 背光控制引脚
31 IO38 SPI_MOSI SPI通信引脚
30 IO37
29 IO36
28 IO35
27 IO0 BOOT BOOT配置引脚
26 IO45 SPI_CLK SPI通信引脚
25 IO48 SPI_CS SPI通信引脚
24 IO47 LCD_DCX 屏幕控制引脚
23 IO21 LCD_Power 屏幕电源控制引脚
22 IO14 AMP_EN 耳机插入检测引脚
21 IO13 I2S0_MCLK I2S通信引脚

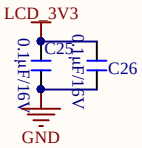

GND
C38
1uF/16V
3V3
U11
PVDD
SDIN
SCLK
LRCK
VMID
VDD
GND
OUTP
OUTN
VRP
ES7111
I2S0_DIN
I2S0_BCLK
I2S0_LRCLK
C39
1uF/16V
C23
4.7μF/25V
C24
4.7μF/25V
R40
470Ω
R42
470Ω
C41
2.2nF/50V
C48
2.2nF/50V
OUTP
OUTN
C49
1uF/16V
C50
1uF/16V
OUTN
OUTP
R39
1K
R44
1K
C42
0.1μF/16V
C46
0.1μF/16V
C40
1uF/16V
AP_EN
R38
1K
R37
100K
TP43
BYPASS1
R41
22K
R43
22K
C45
0.1μF/16V
C47
0.1μF/16V
U10
CTRL
BYPASS
IN-
IN+
OUT-
GND
VDD
OUT+
HT6872
FB5
600Ω/1A
FB6
600Ω/1A
SPK_R-
SPK_R+
5V_OUT
TP16
TP17
C51
22uF/25V
C52
1uF/16V
SPK_R-
SPK_R+
J4
喇叭


A3V3
U12
DVDD
PVDD
AVDD
VRP
VMID
AGND
PGND
DGND
MCLK
SCLK
LRCK
SDIN
SDOUT
HPCOM
ROUTN
LOUTN
LOUTP
ROUTP
C56
1uF/16V
C57
1uF/16V
I2S0_MCLK
I2S0_BCLK
I2S0_LRCLK
I2S0_DIN
TP15
TP19
TP20
TP21
C58
22uF/25V
C59
22uF/25V
R45
33Ω
R46
33Ω
AMP_EN
R31
100K
J5
TP22
TP23
R47
4.7K
R48
4.7K
IIC_CLK
IIC_DATA
CCLK
CDATA
CE
R50
10K
ES8156
R32
910K
TP8
耳机
耳机插入检测


A3V3
R51
10K
R52
10K
R53
10K
J6
TP24
TP25
TP26
SDIO_CMD
SDIO_CLK
SDIO_DAT0
R54
R55
R56
33R
33R
33R
DAT2
CD/DAT3
CMD
CLK
DAT0
DAT1
CD
VDD
GND
GND
GND
GND
GND
GND
GND_SD
R57
1K
C60
0.1µF/16V
A3V3
D12
D13
ESD5302F-3/TR
ESD5302F-3/TR
IO1
IO2
GND
GND
GND
GND_SD
R30
1MΩ
GND


GND
GND
LCD_DCX
SPI_CS
SPI_CLK
SPI_MOSI
LCD_RST
LCD_3V3
LCD_3V3
GND
TP30
TP32
TP31
TP29
J7
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
GND
RS
CS
SCK
NC
NC
NC
SDA
RST
VDD
VDD
A
K
TE
GND
R58
33Ω
Q5
2
3
1
R59
20K
R60
1K
Light_Power
TP18
IRLML2502TRPBF
屏幕接口


GND
LCD_3V3
IIC_DATA
IIC_CLK
LCD_INT
LCD_RST
J8
1
2
3
4
5
6
TP37
LCD_3V3
R15
10K
TP38
GND
GND
触摸屏接口


LCD_3V3
C25
0.1uF/16V
C26
0.1uF/16V
GND